Sicilian, English and Italian Neural Machine Translation

A data pipeline and reproduced baselines, from a small Transformer to a staged NLLB

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Why a new Sicilian translator?

- Sicilian is now covered by massively multilingual models (NLLB-200), but their Sicilian was mined from non-standard web text: about one third of it is dialect, some of it Corsican. The problem moved from the *absence* of data to its *quality*.
- High-quality text exists: since 1979 *Arba Sicula* has published a bilingual Sicilian–English literary journal.
- Wdowiak (2021–22) built the first Sicilian NMT system on it, aligning the sentences by hand.
- This work rebuilds that system on maintained software, with an automatic pipeline, and brings a pretrained model to it.

What is the Sicilian language?

- The Sicilian School of poets at the court of Frederick II created the first literary standard in Italy (13th century), and inspired Dante.
- Dictionaries have presented a consistent orthography for several hundred years (Pasqualino 1785, Mortillaro, Traina, Nicotra).
- It is spoken daily in Sicily, Calabria and Puglia, and in diaspora communities abroad.
- Morphologically rich, with many local varieties and no single official orthography; schools never taught it in writing.
- For NLP: little curated parallel text, much inflection, a literary domain, i.e., a low-resource setting.

Roadmap and contributions

From the baseline to ours:

- data: PDFs to parallel sentences with a sentence encoder;
- baseline: Transformer, subwords, the low-resource recipe;
- pretrained: NLLB-200 adapted with LoRA;
- staged training on scored back-translations;
- results on two test sets, and the baseline retrained.

Contributions:

- an automatic PDF-to-parallel-text pipeline (PyMuPDF + LaBSE), tuned against the hand alignment;
- a curated training set guarded against a frozen literary test set;
- the tokenizer and the desinence-biased segmentation ported to Python;
- NLLB-200 trained in stages, compared with the published systems on their test set and in their tokenized space.

Building the parallel corpus

- Extraction from the PDFs
- Sentence alignment with LaBSE
- Cleaning
- The datasets and the test guard

Feng et al., *Language-agnostic BERT Sentence Embedding* (LaBSE).
Thompson & Koehn, *Vecalign*.

The corpus bottleneck

- *Arba Sicula* sets Sicilian poetry and prose on one page and the English translation on the facing one.
- The text exists only as PDFs, with two columns, captions, front matter and, in the older issues, OCR noise. Wdowiak aligned it by hand with `hunalign` and a Sicilian dictionary.
- Goal: recover the parallel text automatically, so that the dataset can grow without manual alignment.

page $2k$	**page** $2k{+}1$
Sicilian	English
poem / prose	*translation*

Facing pages give a prior on the pairing; the task is to confirm the pairs and to align the sentences within them.

Extraction: from PDF to paired pages

1. Read the text layer with PyMuPDF (no pdflatex round trip). The issues up to AS18, and AS21, are scans without accents, and are kept out of training; from AS19 (1998) the PDFs come from the original files.
2. Assign each page a language by its share of Sicilian and English stop-words.
3. Compare each Sicilian page with the English page on either side and pair them greedily by LaBSE similarity, which corrects mislabelled or swapped pages.
4. Segment into sentences (`pysbd`), drop running headers, page numbers and fragments, and rejoin hyphenated words.

Sentence alignment with LaBSE and a monotonic DP

Embed every sentence with LaBSE (a language-agnostic dual encoder) and build the cosine-similarity matrix

$$S_{ij} = \frac{\langle u_i, v_j \rangle}{\|u_i\| \, \|v_j\|}.$$

Page confirmation: keep the pair if the larger of the mean row and column maxima of S is at least τ_{page}.
Alignment: a monotonic DP over S with moves 1–1, 1–2, 2–1 (gain: the mean cosine) and 1–0, 0–1 (penalty 0.5), banded to the facing page; a span is kept if its mean cosine is at least τ_{sent}.

- No dictionary and no `hunalign`: embeddings and a DP, in the spirit of Vecalign.
- Thresholds chosen on a grid against Wdowiak's hand alignment of AS41–42: $(\tau_{\text{page}}, \tau_{\text{sent}}) = (0.50, 0.40)$, F1 0.58.
- Recall 0.54, precision 0.63: somewhat pessimistic, since a correct pair segmented differently counts as an error.

Cleaning the aligned text

- The errors of the aligner are easy to spot, which makes its output easier to perfect than the one of `hunalign`.
- First the residue of the layout is removed (the InDesign slug of AS43, the "Return to the TOC" links).
- After cleaning and deduplication, the digital archive of the journal gives 15,465 pairs.

A pair is dropped if

1. one side has more than 1.6 times the words of the other;
2. it contains a table-of-contents leader;
3. it has two or more characters typical of OCR noise;
4. more than 25% of its English words are not in a dictionary.

The datasets

- **Web set**: our first extraction (44 issues), Wdowiak's Good-Sicilian-in-NLLB and WikiMatrix it–scn.
- **Curated set**: Wdowiak's hand-aligned sheets, which cover 10–20% of an issue, our alignment of AS19–47 and the Napizia pages.
- **Test guard**: a training pair that contains a sentence of the frozen test set is removed (794 pairs of the curated set).

set	direction	pairs
web	scn–en	27,386
	it–scn	9,389
curated	scn–en	30,980
	scn–it	10,427
	scn (mono)	14,360
test	scn–en	1,000
	it–scn	1,000

The Sicilian data

- Why NLLB's Sicilian is noisy
- The literary standard
- Normalization and its effect

The standard of Cipolla, augmented by Wdowiak for MT.

Why NLLB's Sicilian is noisy

- About one third of the Sicilian collected by NLLB is local dialect, some is Corsican, and few sentences carry a valid translation.
- Multilingual encoders organise around surface form (Verma et al., 2026), so retrieving the *Sicilian* side of an English sentence follows the spelling of the seed data, and Corsican-looking text passes as Sicilian.
- The same proximity to Italian lets our aligner work on a language that LaBSE does not support officially.

LaBSE on FLORES-200 devtest:

scn→en retrieval P@1	99%
Italian (supported)	100%
nearest non-English language	Italian
49%, French 21%	

Alignability follows standardness:

	cosine	$\geq \tau_{\text{sent}}$
standard prose	0.76	91%
dialect poetry	0.64	80%

Standardising Sicilian

- There is no single official orthography, but a literary standard has given a rough consistency for centuries, and most of the Sicilian in NLLB follows it.
- We normalise toward the standard of Cipolla, augmented by Wdowiak for MT: L on articles (*lu/la/li*), H on *aviri* (*haiu, havi, hannu*), CI for *ci*, R kept before a clitic (*farlu*).

Spelling rules ("std" level):

cchiù	→	chiù
io / iu / eu	→	jo
non	→	nun
'un, 'unn	→	nun
pri / pir	→	pi

The "full" level also uncontracts and ASCII-folds.

Normalization: uncontraction, and what it is worth

Uncontract preposition + article:

ô	→	a lu	dû	→	di lu
câ	→	cu la	ntô	→	nta lu

Before: `ntô libbru dû nannu`
After: `nta lu libbru di lu nannu`

The Napizia tokenizer, reimplemented in Python; its output coincides with the one of the Perl code.

Effect on NLLB (scn→en, BLEU / chrF):

	BLEU	chrF
raw	30.67	56.31
std	**30.79**	**56.49**
full	30.61	56.36

"std" is best by only 0.1 BLEU, within the noise; "full" is no better than raw text, as one expects from a tokenizer trained on un-normalized input.

Neural MT: the baseline

- The translation objective
- The Transformer and attention
- Subword segmentation (BPE)
- The low-resource recipe

Vaswani et al., *Attention Is All You Need.*
Sennrich et al., *Neural MT of Rare Words with Subword Units.*
Sennrich & Zhang, *Revisiting Low-Resource NMT.*

The translation objective

Neural MT is a conditional language model: an encoder reads the source $x = (x_1, \dots, x_n)$, and a decoder factorises the target from left to right,

$$p_\theta(y \mid x) = \prod_{t=1}^{|y|} p_\theta(y_t \mid y_{<t}, x),$$

trained by maximising the log-likelihood of the parallel corpus $\mathcal{D}$,

$$\mathcal{L}(\theta) = \sum_{(x,y)\in\mathcal{D}} \sum_{t} \log p_\theta(y_t \mid y_{<t}, x),$$

and decoded with beam search for $\arg\max_y p_\theta(y \mid x)$.

- The Transformer, the subwords and NLLB are different parameterisations of the same p_θ.

The Transformer

No recurrence: tokens interact through scaled dot-product attention. With queries Q, keys K and values V,

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V,$$

computed in parallel heads,

$$\mathrm{MultiHead} = \mathrm{Concat}(\mathrm{head}_1, \dots, \mathrm{head}_h)\, W^O,$$

$\mathrm{head}_i = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V)$. Each layer adds a position-wise FFN, residual connections and layer normalization.

- Self-attention in the encoder and in the decoder; cross-attention lets the decoder look at the source.
- Wdowiak's baseline and NLLB are both encoder–decoder Transformers; they differ in size and in pretraining.

Subword segmentation: BPE

Rare and inflected words are split into subword units, which gives an open vocabulary and shares statistics across the forms of a word. Byte-Pair Encoding:

1. start from characters and count the adjacent pairs of symbols;
2. merge the most frequent pair into a new symbol, and repeat V times.

$$(a, b)^\star = \arg\max_{(a,b)} \ \mathrm{count}(a, b) \quad \Longrightarrow \quad \text{add the token } ab.$$

- Sicilian inflection multiplies the word forms; subwords reduce the sparsity.
- Wdowiak's example: *Carinisi are dogs!* becomes `car@@ in@@ isi are dogs !`

The low-resource recipe

With little data the default Transformer overfits. Sennrich & Zhang (2019) prescribe a smaller and heavily regularised model:

- fewer and narrower layers, a small BPE vocabulary;
- high dropout and label smoothing.

Our baseline uses Sockeye-3 with this geometry, on the CPU.

	default	ours
layers	6	3
model size	512	256
attention heads	8	4
feed-forward	2048	1024

about 6.7M parameters; dropout 0.4 on the embeddings, 0.2 elsewhere; joint BPE of 4,000 merges.

Linguistic levers

Lever B: desinence-biased subwords

- The tokenizer ASCII-folds and uncontracts; the joint BPE is biased toward the textbook desinences (Dieli, Cipolla, Bonner, *Chiù dâ Palora*), which are appended to the Sicilian side when the merges are learned.
- Sockeye baseline 8.51 $\rightarrow$ 10.94 BLEU.

Lever D: lemma source factor

- A lemma embedding is summed into the source embedding (Sennrich & Haddow, 2016),

$$e_j = E_{\text{word}}(w_j) + E_{\text{lemma}}(\ell_j),$$

  with the lemma table of the original system (16,377 forms, 1,252 lemmas).
- It lowers the score to 10.18 BLEU.

Both stay far below the pretrained model of the next section, which reaches 29.00 BLEU without any adaptation.

- NLLB-200
- Parameter-efficient fine-tuning: LoRA
- Bidirectional adapter and back-translation
- The Italian bridge

Pretrained multilingual NMT, adapted

NLLB Team, *No Language Left Behind*.
Hu et al., *LoRA*. Sennrich et al., *Back-translation*.
Johnson et al., *Multilingual NMT*. Fan et al., *Beyond English-Centric*.

NLLB-200: a massively multilingual Transformer

One encoder–decoder trained on 200 languages, Sicilian (`scn_Latn`) among them. The target language is chosen by forcing the first token of the decoder,

$$p_\theta(y \mid x, \ell_{\text{tgt}}), \qquad y_0 = \langle \texttt{scn_Latn} \rangle \text{ or } \langle \texttt{eng_Latn} \rangle.$$

Out of the box it translates literary Sicilian into English at 29.00 BLEU, against 10.94 for the best model trained from scratch.

- We use the 1.3B variant.
- The transfer from 200 languages is what the small model lacks.
- The question is how to adapt 1.3B parameters to some 30,000 pairs on a single GPU.

Parameter-efficient fine-tuning: LoRA

Freeze the pretrained weights $W_0 \in \mathbb{R}^{d\times k}$ and learn a low-rank update (Hu et al., 2021):

$$W = W_0 + \frac{\alpha}{r} BA, \qquad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k},$$

with $r \ll \min(d, k)$; the forward pass is

$$h = W_0\, x \ + \ \tfrac{\alpha}{r}\, B(Ax).$$

- On the query, key, value and output projections, r=32, α=64: 18.9M trainable parameters, 1.4% of the model.
- Base in fp16, adapter in fp32; one Colab GPU.
- The adapter can be merged into W_0 at inference time.

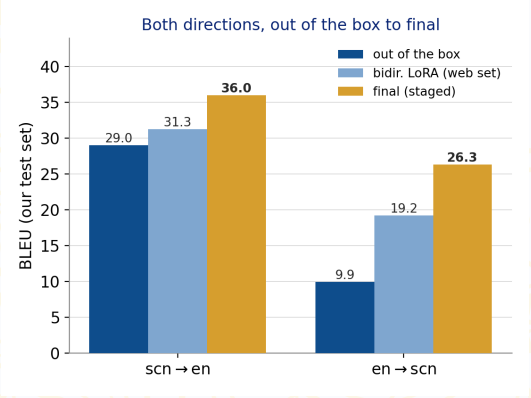
One adapter for both directions

Instead of one model per direction, a single adapter is trained on the union

$$\mathcal{D}_{\leftrightarrow} = \{(x^{\text{scn}}, y^{\text{en}})\} \cup \{(x^{\text{en}}, y^{\text{scn}})\},$$

each example with its forced target token.

- On the web set: scn→en 29.00 → 31.27, en→scn 9.90 → 19.19 BLEU.
- The staged final model reaches 36.01 and 26.34 (next section).



Back-translation and the Italian bridge

Back-translation (Sennrich et al., 2016): with a reverse model $M_{t \to s}$ and monolingual target text,

$$x' = M_{t \to s}(y), \qquad \mathcal{D}^{+} = \mathcal{D} \cup \{(x', y)\};$$

the target side is real, only the source is synthetic.

From the 7,592 English sentences left unaligned: 7,498 pairs, scn→en 31.27 → 31.60. The in-domain pool is small; the staged training uses about a million back-translations instead.

Italian as a bridge (Johnson et al., 2017; Fan et al., 2020): Italian is close to Sicilian and has much more data. A trilingual adapter on scn↔en and it↔scn gives scn→it 43.47 and it→scn 17.57 → 26.97,

but slightly less on scn↔en (30.75 / 17.49 against 31.27 / 19.19): on the web set the bridge does not help the English directions.

- The reverse-training recipe, on NLLB
- Forward-scored back-translations
- The coarse stage
- Ablation

Staged training

From reverse training to a staged NLLB

Wdowiak's recipe: a bidirectional it–en model on ParaCrawl, then Sicilian forward-translations of it, then back-translations of NLLB's Sicilian, then *Arba Sicula*.

Ours: NLLB already provides the multilingual pretraining, so only the last two stages remain.

1. back-translations (6 lists, 4 directions) and 100,000 natural it–en pairs, one epoch;
2. the curated set, scn↔en and scn↔it, two epochs.

Forward scoring: each synthetic pair is scored by the negative log-probability that its real Sicilian side translates the synthetic one, under Wdowiak's models trained on the literary standard; standard Sicilian usually scores lower than dialect.

The best 100,000 pairs per list, after filters: at least 5 words, no near-copies, no test overlap, and few words in -o (Italian passing as Sicilian obtains the best scores).

Staged training: results on our test set

model	scn→en	en→scn
bidir. LoRA (web)	31.27	19.19
staged	32.40	21.72
coarse stage first	33.01	21.78

The gain is larger into Sicilian (+2.53) than from it (+1.13), possibly because the back-translations were selected for the quality of their Sicilian side. The comparison mixes two changes (curated set and back-translations).

The coarse stage (Wdowiak's proposal): one epoch on the 200,000 *worst*-scoring pairs per list, i.e., on the dialectal, Italian and Corsican text, spreads the output probabilities across the dialects; the later stages sharpen them.

It helps from Sicilian (+0.61) but not into it. It also trains one more epoch, so the source of the gain is not certain.

Ablation: what limits the staged model

	base	+ 300k	+ rank 64	+ 3 changes	final
our test (BLEU)					
scn→en	32.82	33.34	34.60	35.40	**36.01**
en→scn	21.52	21.88	24.18	25.09	**26.34**
Wdowiak's test (BLEU, tokenized space)					
en→scn	44.42	44.49	47.64	48.36	**49.25**
scn→en	57.14	58.54	62.02	60.95	59.34

- The three changes: 300,000 pairs per list in the second stage, a rank-64 adapter also on the feed-forward layers, 256-token sequences and a third epoch. Most of the gain comes from the adapter.
- Final: the WikiMatrix it–scn pairs in the third stage and up to four epochs, chosen on the validation set (the best of all variants there).

- Evaluation: BLEU, chrF and the tokenized space
- Our test set
- Forgetting of the untrained directions
- Wdowiak's test set and the retrained baseline

Papineni et al., *BLEU*. Popović, *chrF*. Post, *sacreBLEU*.

Results

Evaluation

BLEU: modified n-gram precision with a brevity penalty,

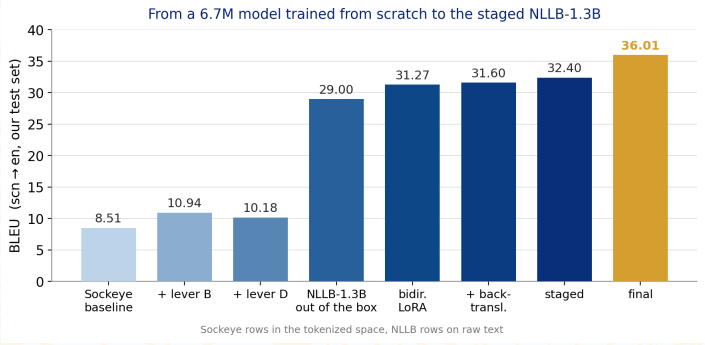
$$\mathrm{BLEU} = \min(1, e^{1-r/c}) \cdot \exp\Big(\textstyle\sum_{n=1}^{4} w_n \log p_n \Big).$$

chrF: the F-score of character n-grams, kinder to rich morphology,

$$\mathrm{chrF}_\beta = (1+\beta^2)\, \frac{\mathrm{chrP} \cdot \mathrm{chrR}}{\beta^2\, \mathrm{chrP} + \mathrm{chrR}}.$$

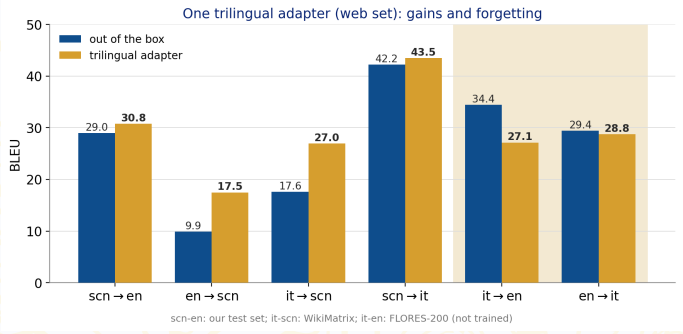
- sacreBLEU (13a) on our frozen test set of 1,000 literary pairs; not FLORES, whose Sicilian follows the post-2017 orthography and another domain.
- Wdowiak's test set: 121 trilingual lines, scored as he did in the space of the Napizia tokenizer (accents folded, contractions undone).
- That space is more lenient into Sicilian: en→scn 21.72 → 25.69, scn→en 32.37 → 33.03.

Our test set (scn→en)



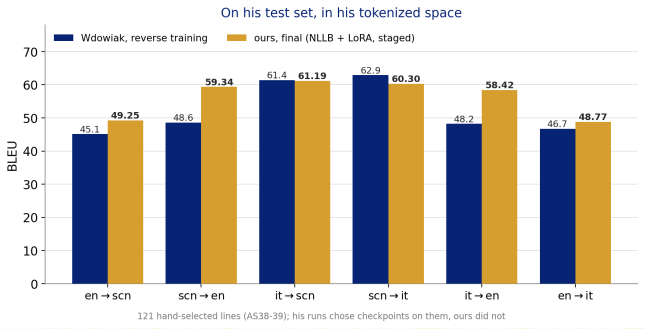
- The Sockeye baseline is 3.3 BLEU above the floor of the untranslated source (8.51 against 5.18).
- NLLB out of the box is 18 BLEU above the best model trained from scratch.
- The staged training and the ablation bring it to 36.01 (26.34 into Sicilian).

Forgetting of the untrained directions



- The trilingual adapter lowers it→en from 34.42 to 27.14 BLEU on FLORES-200, a direction it was never trained on.
- This limits its use as a general multilingual model; Sicilian to and from Spanish, French or Portuguese is probably affected too.

Against the published systems



- Final model against Wdowiak's reverse training: +4.2 BLEU into Sicilian, +10.7 from it, and above it between English and Italian.
- Below it between Sicilian and Italian (61.19 against 61.4, 60.30 against 62.9).
- Not entirely even: his systems chose their checkpoints on these lines, ours train for a fixed number of epochs.

Hence, the low en→scn score on our test set is probably due to the test set and to the metric more than to the model.

The original baseline, retrained

training set	pairs	en→scn	scn→en
published	–	25.1	29.1
A (his sheets)	20,220	15.97	16.65
B (our alignment)	18,503	13.20	13.14
C (both)	31,730	22.45	20.69

His test set, his tokenized space, BLEU.

- His 2022 configuration (Napizia tokenization, separate BPE of 3,000 merges, 3 layers of size 256), with Sockeye 3 instead of Sockeye 1.
- On his data it stays well below the published scores: he started each model from a previous one and stopped on the test set; none of our runs converged in 20 epochs.
- Our alignment added to his sheets raises it by 4.0 to 6.5 BLEU: the automatic alignment brings pairs that the hand alignment does not contain.

Conclusion and outlook

- The path from the PDFs of *Arba Sicula* to a Sicilian–English system was rebuilt on maintained software: extraction, alignment and cleaning, the tokenizer ported to Python, and NLLB-200 with LoRA trained in stages.
- On our test set the final model reaches 36.01 BLEU (scn→en) and 26.34 (en→scn); on Wdowiak's, it is above his reverse-training system in the directions that involve English and below it between Sicilian and Italian.
- The model also runs on the CPU: converted to CTranslate2 with int8 weights it takes 1.4 GB on disk and 2.5 GB of memory, about 3 s per sentence with 2 threads, and loses less than 0.5 BLEU.
- Next: limiting the forgetting with a multi-parallel corpus, and the same pipeline for Calabrese, Salentino and the other varieties of Italy (LaBSE retrieves their English translation in 98.5–99.9% of the cases on FLORES-200).

Paper: arXiv:2610.30374. Code: `github.com/dmeoli/SicilianNMT`.

Acknowledgements

- **Arba Sicula**, **Gaetano Cipolla** and **Arthur Dieli**, for the bilingual journal, the vocabulary and the grammar, and for the permission to use their digital collections.
- **Eryk Wdowiak**, co-author of the paper, whose *Tradutturi Sicilianu*, hand-aligned sheets and back-translations this work builds on.
- The open-source projects it stands on: PyMuPDF, LaBSE and sentence-transformers, Sockeye, NLLB-200, PEFT, sacreBLEU, CTranslate2.

Grazzii assai.

Grazzii *p'a vostra attenzioni*!

github.com/dmeoli/SicilianNMT